

Limiting the Uncoordinated N Species in $M-N_x$ Single-Atom Catalysts toward Electrocatalytic CO_2 Reduction in Broad Voltage Range

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Carbon-supported single-atom catalysts (SACs) are extensively studied because of their outstanding activity and selectivity toward a wide range of catalytic reactions. Amidst its development, excess dopants (e.g., nitrogen) are always required to ensure the high loading content of SACs on the carbon support. However, the use of excess dopants is accompanied by formation of miscellaneous structures (particularly the uncoordinated N species) on catalysts, leading to adverse effects on their performance. Herein, the synthesis of carbon-supported Ni SACs with precisely controlled single-atom structure via joule heating strategy, showing the coordination of 80% of N dopants with metal elements, is reported. The preclusion of the unfavorable N species is confirmed to be the main reason for the superior performance of optimized Ni SACs in electrocatalytic carbon dioxide reduction reaction, which demonstrates unprecedented activity, selectivity, and stability under an exceptionally broad voltage range (>92% CO selectivity in the range of -0.7 to -1.9 V reversible hydrogen electrode). Such a synthetic strategy is further applicable for the design of SACs with various metals. This work demonstrates a facile method for preclusion of unfavorable dopants in the SACs and its importance in catalytic application.

are extremely reactive, metal centers are commonly stabilized on supports via coordination of nonmetal dopants. In this regard, excess dopants are employed to optimize the loading content of SACs on the carbon. Such an excess content of dopants can in turn lead to inevitable interference on the catalytic performance of SACs,^[2] which may hamper the mechanism study on the catalytic reaction. Taking nitrogen (N, the most commonly explored dopant for preparation of carbon-based SACs) for instance, with various metallic loading of single-atom catalysts,^[3] most of the N dopants do not take part in coordinating with SACs through formation of $M-N_x$ species (M, metallic atom and x , coordination number). The other N dopants remained in the form of pyrrole N, pyridine N, graphitic N, and N–O on the carbon-based SACs. During the catalytic reaction, these N species can possibly induce various unfavorable effects on the carbon-supported SACs. In many cases,

enormous property differences have been observed on the SACs that were reported to have similar or even the same structure, concluding contradict relationship between coordination and selectivity.^[4] For example, similar $Ni-N_x$ SACs showed different potential window of CO selectivity in electrocatalytic CO_2 reduction reaction (CO_2RR).^[1c,5] As a consequence, it remains a huge gap between the understanding on the correlation of the

1. Introduction

In the past several decades, carbon-supported single-atom catalysts (SACs) with various isolated atoms and coordination structures have been extensively studied owing to their unique features of exclusively high catalytic activity and effective utilization of metal centers.^[1] Considering that the SACs

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SAC structures with their catalytic performance. These paradox phenomena give rise to an important challenge: is it possible for us to synthesize a clearly structured SAC that has only desired single-atom active center on the highly graphite carbon without much extra defects?

To explore this possibility, we choose a joule heating device to synthesize the $M-N_x$ SACs controlling coordination structures. The joule heating strategy featuring high maximum temperature, ultrafast heat, and quenching, has recently attracted much attention from scientific community. Specifically, through the joule heating device, a target material can be heated up to ≈ 3000 K within a few milliseconds. Such an instantly elevated temperature opens up a wide range of opportunities for preparation of various materials such as graphite carbon,^[6] metal nanoparticles,^[7] and complex metal oxides.^[8] Therefore, it can be expected that such a heat shock method can introduce magnificent benefits for preparation of carbon-supported SACs. For example, Yao et al. reported that the high temperature shock-wave induced by joule heating strategy can be utilized for preparation of Pt SACs supported by defect carbon with superior stability.^[9] Herein, we report the synthesis of carbon-supported $Ni-N_x$ SACs via joule heating by employing the metal-ligand complex adsorbed by carbon black as the precursor. By selecting 1,10-phenanthroline (Phen)- $Ni(OAc)_2$ (acetate) as precursor, the majority (up to 80%) of N dopants in the obtained material

are utilized to coordinate with the metal centers, reducing the unfavorable uncoordinated N species in the Ni SACs (Table S1, Supporting Information). To demonstrate the superiority of the SACs prepared via this method, we employ electrocatalytic CO_2RR —a highly important energy-related reaction as a model case. The optimized $Ni-N_x$ demonstrates an excellent CO selectivity of $\approx 92\%$ throughout the applied voltage range of -0.7 to -1.9 V versus reversible hydrogen electrode (RHE) in an H-cell with 0.1 M $KHCO_3$ as electrolyte. It should be noted that its current density toward CO can achieve 70 mA cm^{-2} at an extremely negative applied voltage of -1.9 V in an H-cell. More importantly, the stability limitation of the SAC is overcome in this work, in which the optimized sample demonstrates a stable function (activity loss less than 5%) for 48 h of continuously working under -1.5 V. Furthermore, such a method for preparation of fully coordinated metal- N_x can be extended to a variety of metal centers.

2. Results and Discussion

In this work, our model catalyst is prepared using our homemade joule heating system (Figure 1a). During the preparation, the specific content of precursors for nickel ($Ni(OAc)_2$) and nitrogen (1,10-phenanthroline) is first adsorbed by the

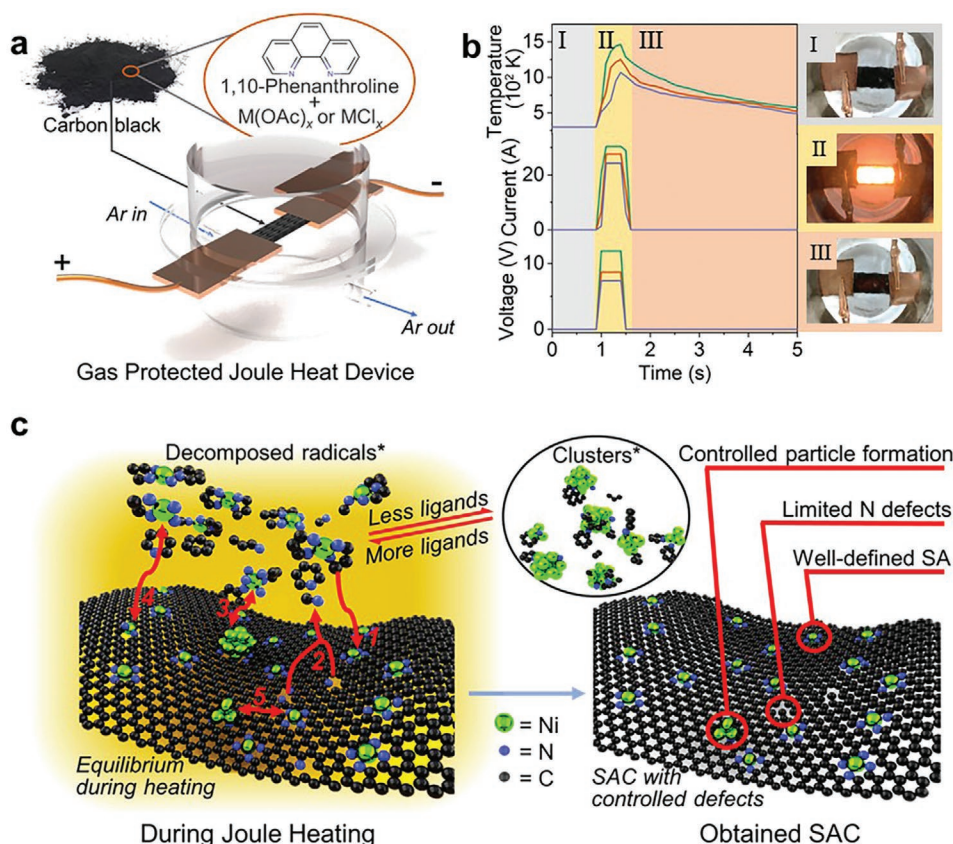


Figure 1. a) Schematic illustration of the homemade reaction system for joule heating. b) Temperature, direct current, and voltage changes with the evolution of joule heating duration (left), and optical images before, during, and after heating of the samples on reaction platform (right). c) Schematic illustration of joule heating processes for preparation of Ni SACs.

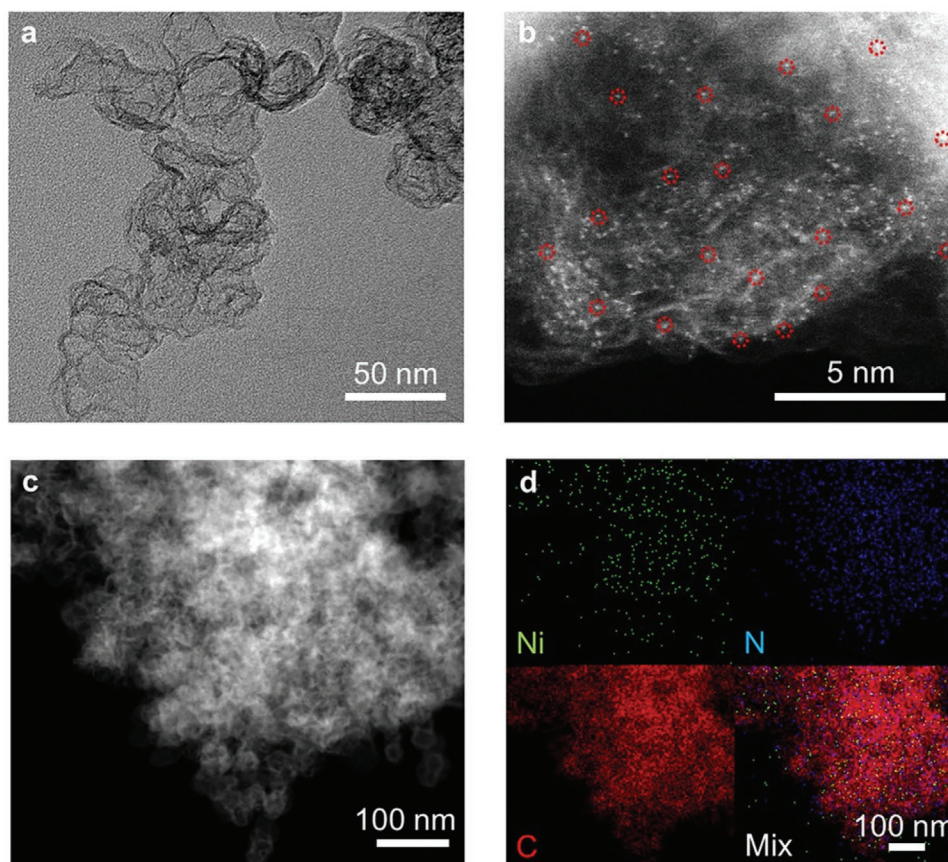


Figure 2. a) HRTEM, b) HAADF-STEM, c) STEM, and d) EDX elemental mapping profiles for Ni, N, C, and all elements of C250Ph3.5Ni SAC.

carbon black. The applied voltage and current are set to ensure that an instant heat of 1300 °C can reach the sample in less than 0.5 s (Figure 1b), which is beneficial for optimizing the physicochemical properties of the prepared samples (see Figures S1 and S2, Supporting Information). Finally, the samples can be obtained and labeled as $C\alpha Ph\beta Ni\gamma$ according to their molar ratio of precursors. To have a clear image for this joule heating strategy, we illustrate the mechanism of this process in Figure 1c. Generally, the joule heating process for carbon-supported SACs preparation can be divided into two main steps. First, $Ni-N_x$ radicals resulting from the interaction of the evaporated Ni and N of precursors^[10] are captured by the defects on carbon black substrate. Second, the thermodynamically unfavored metal-free N species are decomposed and precluded from carbon black substrate under high temperature.^[6] Apart from these two steps, another three factors should be taken into account during the joule heating process. The excess concentration of the ligands should be always employed to guarantee the high dispersion of the metal centers for formation of SACs on the carbon black.^[9a,11] If the concentration of the ligands is insufficient, the metal centers tend to accumulate to form metal clusters which are unfavorable for maximizing the surface active sites of the metal catalyst. In addition, another critical issue is that the unstable metal centers can either evaporate to atmosphere or capture vapor phase ligand from the atmosphere to form stable single-atom sites on carbon black.^[5b] Depending on the thermodynamic

stability of the metal centers, they can either aggregate to form larger particle or disperse to form single atoms.^[9a,12]

To confirm that the carbon-supported Ni SACs are successfully prepared, we then examine the structural properties of the obtained Ni SACs. X-ray diffraction (XRD) (Figures S1–S3, Supporting Information) shows that all the prepared samples demonstrate two main peaks at 24° and 44° attributed to the (002) and (100) planes of the graphite. In addition, according to the Raman spectroscopy characterization, the carbon support is highly graphitized (Figure S4, Supporting Information). Specifically, two distinct peaks at 1359 and 1595 cm^{-1} are assigned to sp^3 disorder carbon (D band) and sp^2 graphitic carbon (G band), respectively. After the joule heating, the I_D/I_G ratio of the Ni SACs (1.15) is significantly reduced compared to that of the pristine carbon black (1.50), indicating the increased graphitic content of the substrate.^[5a] Furthermore, the morphology of the Ni SACs is investigated via transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM), showing that the structure of the carbon black is preserved after the joule heating (Figure 2a). By performing aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM), the homogeneously distributed bright dots can be observed on the Ni SACs, demonstrating the existence of single atoms (Figure 2b). The elemental mapping characterization by scanning transmission electron microscopy (STEM) equipped with energy-dispersive X-ray (EDX) mapping device (Figure 2c,d), further

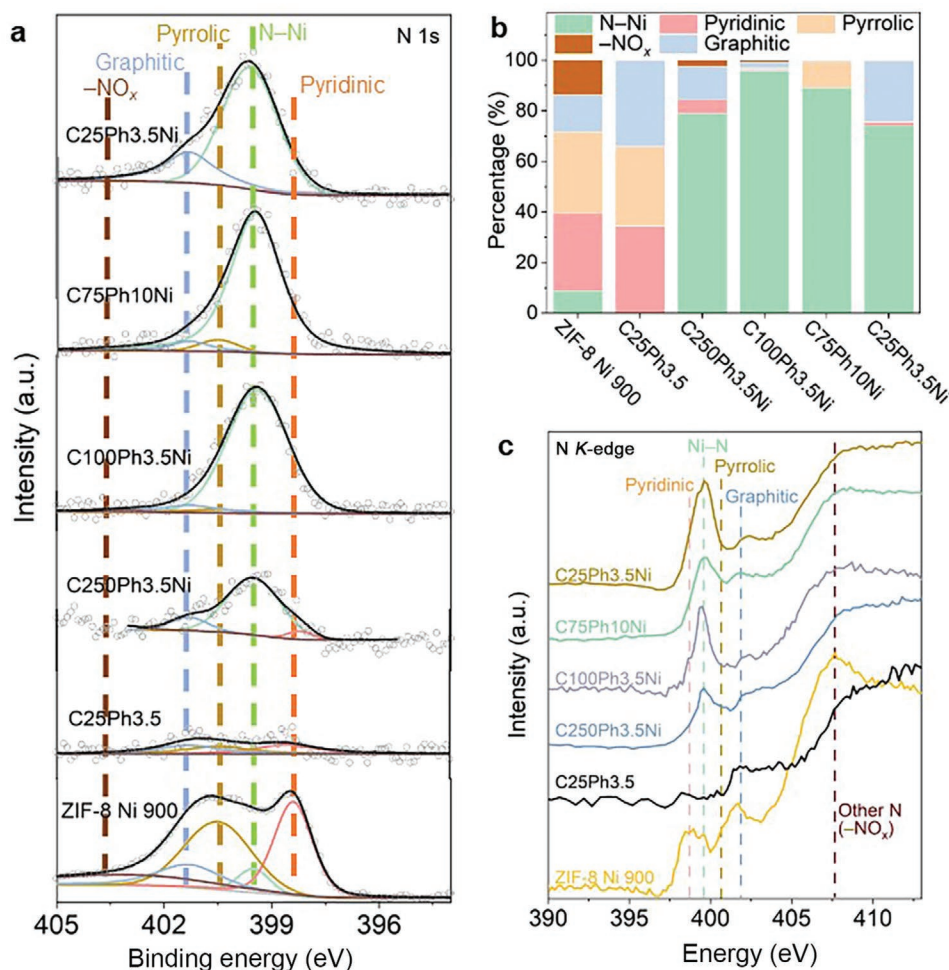


Figure 3. a) N 1s XPS spectra, b) N composition derived from N 1s XPS, and c) N K-edge XAS spectra for C25Ph3.5, C250Ph3.5Ni, C100Ph3.5Ni, C75Ph10Ni, C25Ph3.5Ni, and ZIF-8 Ni 900.

confirms that these single atoms are attributed to the Ni elements, showing the homogeneous dispersion of Ni and N over the substrate. The mass loading of Ni is determined by inductively coupled plasma-atomic emission spectrometry using a calcination–dissolution strategy, indicating a loading of 1.0 wt% (Table S1, Supporting Information).

To clarify the status of the N dopants in the NC-SACs, we perform the X-ray photoelectron spectroscopy (XPS) characterization. Here the traditional Ni SAC by annealing metal–organic framework is employed as a reference sample to assign the XPS spectra. We take 9% Ni-doped zeolitic imidazolate framework-8 (ZIF-8) calcinated at 900 °C (i.e., ZIF-8 Ni 900) as an example. As shown in Figure 3a, the XPS N 1s spectrum of ZIF-8 Ni 900 can be deconvoluted into pyridinic N (≈ 398.3 eV), Ni–N (≈ 399.5 eV), pyrrolic (≈ 400.5 eV), graphitic N (≈ 401.3 eV), and oxidized N (≈ 403.0 eV).^[13] In sharp contrast, our samples generated via joule heating are dominated by Ni–N signal. Generally, the peaks attributed to the pyridinic N and pyrrolic N can be significantly reduced by increasing Ni precursor. For all the C α Ph β Ni γ samples, the N species other than the Ni–N are almost eliminated, suggesting that we have successfully prepared the Ni SACs with minimized N species. To clearly

illustrate such a minimized N content in our sample, we estimate the contents of different N species on the sample based on the obtained XPS results. As shown in Figure 3b, only $\approx 10\%$ of N dopants in the ZIF-8 derived Ni SACs are utilized for stabilizing the Ni atoms. In comparison, all the Ni SACs prepared through joule heating strategy demonstrate high proportion of Ni–N ($>80\%$). It is worth noting that Ni and N atoms are simultaneously implanted in carbon black during joule heating. Without the presence of Ni atoms, N species are barely observed. Out of caution, we also perform the X-ray absorption spectroscopy (XAS) for N elements (Figure 3c). The results are in good accordance with the XPS characterization, that the C250Ph3.5Ni shows almost only the Ni–N species with minor graphitic N, pyrrolic N, and pyridinic N.^[14] Combining the XPS and XAS characterizations, it can be confirmed that joule heating can be a superb strategy to get rid of the unfavorable N species on the Ni SACs.

To resolve the local structural information for Ni atoms, the samples are further characterized by synchrotron radiation-based X-ray absorption fine structure (XAFS) spectroscopy. Figure S5 of the Supporting Information shows the Ni K-edge X-ray absorption near-edge structure (XANES) spectra for the

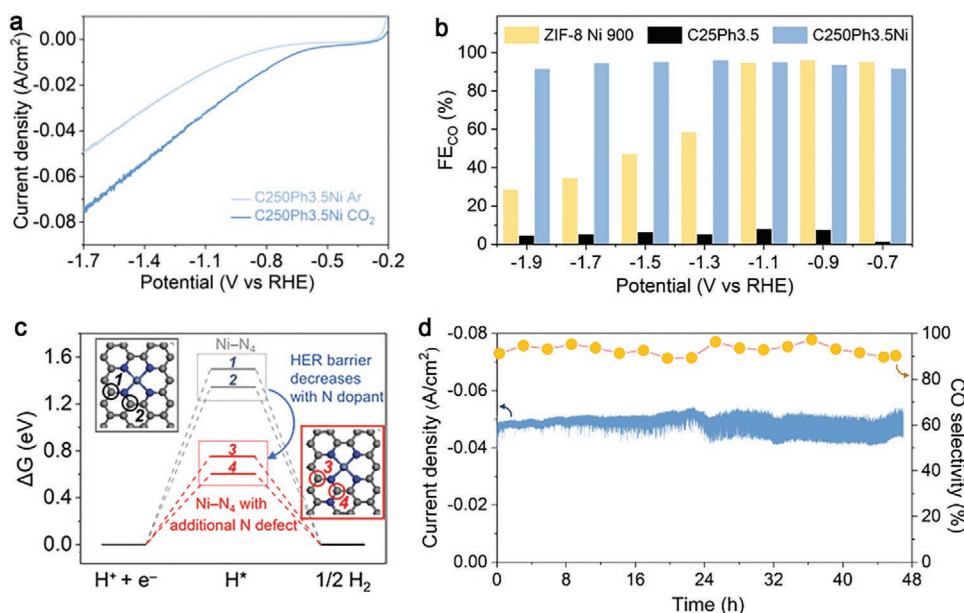


Figure 4. a) LSV curves and b) faradaic efficiency for CO production of different prepared samples. c) Gibbs free-energy profiles for HER on substrate containing NiN₄ or NiN₄ with additional graphitic N dopant. d) Stability test using C250Ph3.5Ni SAC under -1.5 versus RHE.

CαPhβNiγ and ZIF-8 Ni 900 in reference to standard Ni foils. Processed through a Fourier-transform (FT), the Ni K-edge extended XAFS (EXAFS) spectra are obtained (Figure S6, Supporting Information). To obtain the local bonding information for our CαPhβNiγ sample, we extract bond lengths and coordination numbers from EXAFS curve fitting. Consistent with the XRD results, the ratio of Ni–Ni and Ni–N bonds can be adjusted with different molar ratio of carbon defects, N precursor, and Ni precursor. As listed in Table S2 of the Supporting Information, the Ni K-edge spectrum reveals that the coordination number of Ni–N in C250Ph3.5Ni is 3.37. This indicates that Ni atoms in our samples are mostly coordinated with N.

Such materials obtained under high-temperature joule heating without miscellaneous substrate structure will open a new insight of looking into carbon-supported SAC. To this context, we take electrocatalytic CO₂RR as a model reaction to study the possible function of our prepared SAC in catalytic reaction (Figure 4; Figures S7–S10 and Table S3, Supporting Information). Electrochemical tests are performed in CO₂ saturated 0.1 M KHCO₃ using a H-cell and Nafion as a proton exchange membrane. The performances of the prepared samples are tested on carbon paper with a loading content of 0.5 mg cm⁻². As shown in Figure 4a, linear sweep voltammetry (LSV) curves performed in Ar (C250Ph3.5Ni Ar) and CO₂ (C250Ph3.5Ni CO₂) saturated electrolytes reveal that the catalysts possess the activity for CO₂RR. To evaluate the reaction selectivity of catalysts, we compare the faradaic efficiency for CO₂RR to CO (FE_{CO}) of prepared samples at different potentials (Figure 4b). Clearly, the optimized Ni SAC (C250Ph3.5Ni) demonstrates a high FE_{CO} over a wide range of applied voltages (-0.7 to -1.9 V vs RHE), suggesting its superior selectivity toward CO production. In addition, no liquid products can be detected in the aqueous phase by H¹ nuclear magnetic resonance spectroscopy, further confirming its high selectivity to CO. Such an FE_{CO} of Ni SAC is higher than that of the most state-of-the-art

catalysts (Table S4, Supporting Information) under high potential (<-1.2 V) in H-cell. For example, the ZIF-8 derived Ni SAC (ZIF-8 Ni 900) demonstrates a low FE_{CO} at the range of -1.3 to -1.9 V versus RHE (Figure 4b). This result is because the presence of abundance unfavorable N species, especially pyridinic N or graphitic N,^[15] can provoke the competing HER at a relatively negative applied potential, thereby suppressing the CO₂RR performance of the Ni SAC. In addition, the other samples with more Ni–Ni existing but few uncoordinated N, prepared through joule heating with different Ni content (C100Ph3.5Ni and C75Ph10Ni), exhibit a plausible FE_{CO} (>90% throughout -0.7 to -1.9 V, Figure S7, Supporting Information), further confirming the positive effect of limiting extra uncoordinated N species on the carbon-supported Ni SAC.^[1c,5c] In comparison, the samples are prepared using the same precursors as those for C75Ph3.5Ni, C100Ph3.5Ni, and C250Ph3.5Ni via heating in a tube furnace at 1300 °C for 1 h with a heating rate of 2 °C min⁻¹ instead of joule heating, which are named as C75Ph3.5Ni-TF, C100Ph3.5Ni-TF and C250Ph3.5Ni-TF, respectively. As revealed by XRD characterization (Figure S11, Supporting Information), some impurity phases can be found on C250Ph3.5Ni-1300TF, which are detrimental to the catalytic performance. In addition, Ni–Ni peak is found dominant in the XAFS spectra of these samples (Figures S12 and S13, Supporting Information), suggesting that the formed Ni species on the samples prepared by traditional heat treatment are mostly aggregated. The FE_{CO} of these samples prepared by traditional heat treatment (Figures S14 and S15, Supporting Information) is far lower than that by joule heating strategy in electrocatalytic CO₂RR due to the presence of the metallic Ni. These results demonstrate that the joule heating strategy is an ideal strategy for the preparation of the SACs with minimized uncoordinated N and crystallized Ni.

To investigate the influence of the N species on the CO₂RR performance of the Ni SACs, we perform the first-principles

density functional theory simulation.^[15] According to the EXAFS fitting results, we first build different structures of the Ni SACs including single Ni–N₄ site and Ni–N₄ site with an extra graphitic N species,^[16] and calculate their corresponding energy for HER. Generally, the ΔG for HER on the carbon atoms close to the Ni–N₄ site is calculated to be 1.345 (site 1) and 1.498 (site 2) eV (Figure 4c). After adding the graphitic N species to the structure, the ΔG for HER is significantly reduced to 0.603 (site 1) and 0.753 (site 2) eV, suggesting the enhanced HER activity on N-doped substrate. Similar results can also be obtained when adding extra pyrrolic N or pyridinic N species to the Ni–N₄ site (Figures S16 and S17, Supporting Information). Such results show that the presence of N species other than Ni–N_x provokes the HER and the nearby graphitic N suppresses the CO₂RR. Apart from the uncoordinated N species around the coordinated N, we have also studied HER activity of the graphitic N, pyridinic N, and pyrrolic N which are separated from the coordinated N. As shown in Figures S18 and S19 of the Supporting Information, the ΔG values of graphitic N, pyridinic N, and pyrrolic N are far lower than that of the Ni–N₄, suggesting that the uncoordinated N can greatly enhance the HER on the catalysts, which is consistent with previous HER research on metal-free N-doped carbons.^[15,17] According to the calculation result, another comparative experiment is performed based on C250Ph3.5Ni with extra N depositing (Figure S20, Supporting Information). N 1s XPS and N K-edge XAS spectrum of C250Ph3.5Ni-extra N show the increased concentration of graphitic, pyridinic, pyrrolic, and oxidized N species. More importantly, Ni K-edge XAFS characterization of C250Ph3.5Ni-extra N confirms that the coordination number of Ni–N for C250Ph3.5Ni-extra N is similar to that of C250Ph3.5Ni. Such results provide us a perfect platform for investigating the influence of uncoordinated N on catalytic performance. With additional uncoordinated N species, the selectivity of CO production decreases with generation of H₂. This result demonstrates that the preclusion of the unfavorable N species through the joule heating strategy plays the major role in enhancing the catalytic performance of the prepared Ni SACs in CO₂RR by suppressing the competing HER under a very negative voltage.

Many carbon-supported SACs have a limited stability. Therefore, we study the long-term performance of the prepared Ni SAC for the CO₂RR to evaluate its electrocatalytic stability. As shown in Figure 4d, current density remains stable with a negligible decrease in FE_{CO} after reaction for 48 h with an applied voltage of –1.5 V versus RHE, suggesting its perfect stability. XRD (Figure S21a, Supporting Information), N 1s XPS (Figure S21b, Supporting Information), and HADDF-STEM (Figure S22, Supporting Information) characterizations are performed for the C250Ph3.5Ni after the electrochemical tests. No obvious changes can be found for C250Ph3.5Ni before and after the electrochemical tests via all these characterizations. These results conclusively confirm that the C250Ph3.5Ni owns good stability for the electrocatalytic CO₂RR. To further demonstrate the robustness of the prepared Ni SAC, we treat the Ni SAC in aqua regia, which can oxidize unstable carbon regions to dissolve the neighbored Ni SAs, for 3 days. After the treatment, the CO₂RR performance of the Ni SACs is mainly preserved and its FE_{CO} and corresponding current density are negligibly changed in the applied voltage range of –0.7 and –1.3 V RHE

(see Figure S23, Supporting Information). The selectivity drop could result from the N-contained groups introduced during aqua regia treatment. This fascinating stability of the Ni SACs is enabled by the fact that only Ni–N_x with exceptional thermodynamic stability can be remained on the prepared sample after the joule heating. Furthermore, we perform the flow cell electrocatalytic CO₂RR using our optimized sample (C250Ph3.5Ni, Figures S24–S28, Supporting Information). As shown in Figure S27 of the Supporting Information, the C250Ph3.5Ni can reach a high current of 600 mA using a 1 cm² gas diffusion electrode (GDE) with a CO selectivity over 95%. Moreover, by comparing the SEM images of the electrodes before and after the electrochemical test (Figure S28, Supporting Information), no obvious detachment of the sample from the electrode and destruction of the sample structure can be observed, confirming the high stability of the prepared GDE. Another joule heating processed material (C100Ph3.5Ni) with more Ni loading but limited uncoordinated N is also tested in flow cell to confirm the benefits of limiting extra uncoordinated N species. Based on these CO₂RR tests, joule heating can not only largely limit the uncoordinated N species on Ni SACs for enhancing its CO₂RR performance, but also guarantee the robustness of stably coordinated Ni–N_x for strengthening its electrocatalytic stability.

For the sake of demonstrating the universality of this joule heating method for preparation of carbon-supported SACs, we prepare a series of SACs with different metals including Fe, Co, Cu, and Zn. Similar to the Ni SACs, all the prepared SACs show only the XRD peaks attributed to the graphite (Figure 5a), and their Raman spectra demonstrate the well crystallized graphite structure (Figure S28, Supporting Information). The N K-edge XAS and N 1s XPS characterization further suggest that most of the existing N elements on the substrate are coordinated with the metal elements instead of forming unfavorable N species (Figure 5b,c). In addition, the HADDF-STEM images indicate that the metal atoms can be homogeneously distributed on the carbon support irrespective of the types of metal elements (Figure 5d–g). EXAFS results show that the metallic bonds are scarce on all these SACs, suggesting the successful preparation of the single-atom catalysts (Figures S30 and S31, Supporting Information). In addition, we have modified the setup of our joule heating device for continuous production of the SACs (Figure S32, Supporting Information). Specifically, the carbon clothes with the precursors can be continuously fed into the heating region of our joule heating device, incessantly producing the target SACs. These results clearly show that the joule heating strategy is a universal method for preparation of carbon-supported SACs and is possible for the usage of large-scale production.

3. Conclusion

In summary, we demonstrate that the joule heating is an excellent strategy for preparation of carbon-supported SACs with minimized N dopants. Employing the electrocatalytic CO₂RR reaction as a proof-of-concept example, it is confirmed that the identification and controlling of miscellaneous N species, such as graphitic N, pyrrolic N, and pyridinic N, can be critical for optimizing the catalytic performance of such carbon-supported

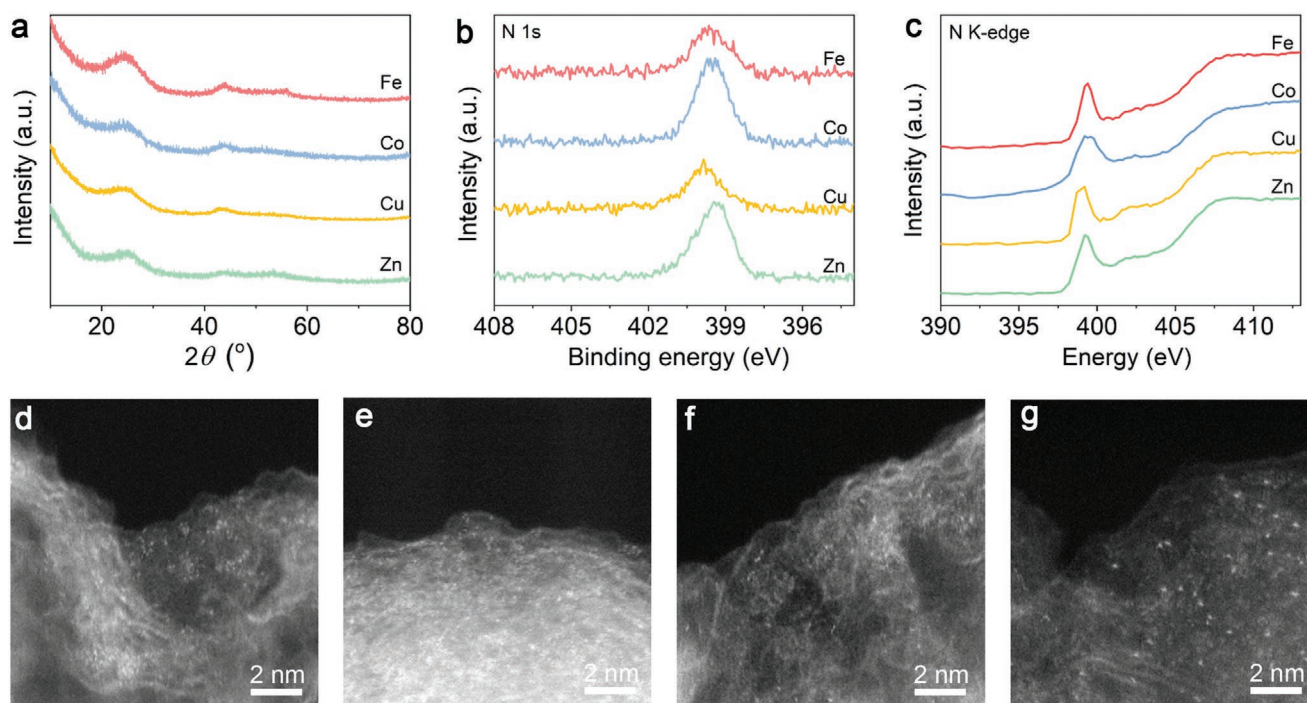


Figure 5. a) XRD patterns, b) N K-edge XANES spectra, and c) N 1s XPS spectra of Fe, Co, Cu, and Zn SACs prepared via joule heating. HADDF-STEM images of d) Fe, e) Co, f) Cu, and g) Zn SACs.

SACs. The absence of these unfavorable N species in Ni SACs can suppress the competing HER during the CO₂RR, and thus the optimized Ni SACs exhibit a remarkably high FE_{CO} over a wide applied voltage range (−0.7 to −1.9 V vs RHE) during the CO₂RR. More importantly, most of the remained N dopants are coordinated with the Ni elements, which are thermodynamically stable, allowing the Ni SACs to have a high electrocatalytic and chemical stability. Such a splendid joule heating strategy has been also expanded for preparation of carbon-supported Fe, Co, Cu, and Zn SACs, showing its universality for preparation of SACs. This work provides new insights for the preparation and interpretation of carbon-supported SACs toward various catalytic applications.

4. Experimental Section

Materials and Device Set-Up: All chemical reagents were purchased from Shanghai Chemical Reagent Factory (China) without further purification. Deionized (DI) water was used throughout this work. Joule heating was performed using a homemade reactor (see Figure 1a). The plate was utilized as connector to fix the reaction platform for joule heating. During joule heating, Ar flowed through the device, direct current was applied to the carbon clothes by linking it to copper plate. Carbon clothes were cut into rectangle with a size of 10 × 40 mm². Then the carbon clothes were cleaned by heating it to 2000 °C for 5 s and repeated for five times, which can then be readily used as the reaction platform for joule heating.

Preparation of C250Ph3.5Ni SAC: 8.3 mg nickel(II) acetate tetrahydrate and 23.2 mg 1,10-phenanthroline monohydrate (C₁₂H₈N₂·H₂O) were dispersed in 50 mL ethyl alcohol and stirred for 10 min with a molar ratio of 1:3.5. Then 100 mg carbon black (Ketjen Black) was added with a Ni:C molar ratio of 1:250, followed by stirring and ultrasonication for 20 min. After that, the precursor-modified carbon black was dried and ground.

Subsequently, the obtained black mixture was put between carbon clothes for joule heating treatment under a 1300 °C peak temperature for 0.5 s and repeated five times under Ar flow. The obtained sample was denoted as C250Ph3.5Ni. Reference samples (CαPhβNiγ) were prepared by changing the precursor composition and labeled according to their molar ratio of precursor composition.

Preparation of ZIF-8 Derived SACs: ZIF-8 derived N-doped carbon-supported SAC was synthesized by carbonizing the 9% Ni contained ZIF-8 in 900 °C tube furnace with a heating rate of 5 °C min. Finally, the ZIF-8 derived SACs can be obtained by washing and ultrasonication in concentrated HCl for 8 h. The obtained sample was denoted as ZIF-8 Ni 900.

XAFS Characterization: Metal K-edge XANES and EXAFS data were obtained from the beamline BL14W1 of the Shanghai Synchrotron Radiation Facility and 1W1B in Beijing Synchrotron Radiation Facility. The electron beam energy of the storage ring was 3.5 GeV with a maximum stored current of 300 mA. Data processing and analysis were performed following standard methods. In detail, the EXAFS data were isolated using the Athena and Artemis software.^[18] The normalized EXAFS was then Fourier transformed over a photoelectron momentum (*k*) range of 2.0–12.5 Å^{−1}. The coordination shells to be analyzed were isolated by inverse transforming over a non-phase-corrected radial distance (*R*) range of 0.7–2.7 Å. Structural parameters were determined by nonlinear least square fitting using IFEFFIT, with phases and backscattering amplitudes calculated ab initio with the FEFF8.1 code. The amplitude reduction factor (*S*₀²) was determined from bulk metallic Ni/NiO/nickel(II) phthalocyanine (NiPc) standard and held constant for the analysis of the samples.

XAS Characterization for N Element: The N K-edge XAS data were collected at the MCD Endstation of the BL12B-A Soft X-Ray beamline at the National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. All XAS spectra were processed and normalized using the Athena software program.^[18]

Electrocatalytic CO₂ Reduction Measurement: Electrodes for CO₂ electrolysis were prepared by casting the catalysts-isopropanol-Nafion solution (5 mg mL^{−1} catalysts, 0.15% Nafion) on hydrophilic carbon paper (0.25 cm²) or glassy carbon electrode with a radius of 2.5 mm.

The catalysts loading was set to 0.5 mg cm². The electrocatalytic CO₂RR was conducted in a gas-tight H-shape electrochemical cell at different potentials. Electrolyte was bubbled with CO₂ or Ar for 20 min before the experiments. Cyclic voltammetry was first performed to stabilize the catalysts. Electrocatalytic activities of the prepared samples were evaluated for electrochemical reaction in CO₂ or argon saturated 0.1 M KHCO₃ solution using LSV with a scan rate of 5 mV s⁻¹. The selectivity of the CO₂RR was tested using current-time mode under a constant applied voltage. The products of the reaction were then investigated by gas chromatography when the quantity of electric charge was around or exceeded 5 C.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

CO₂ reduction, electrocatalysis, joule heating, N species, single atom

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